

David Hughes's Work Explained Simply

PART 1: The Accuracy and Appropriateness Model of Validation

What it is and why Hughes created it

For most of the last century, the word "validity" in psychometrics (the science of measuring human traits through tests and questionnaires) has been used as a catch-all term that tried to mean everything at once. By the 2010s, the field's official guidance said validity was a single unified concept covering about five different types of evidence: everything from whether a test measures what it claims to measure, all the way to whether using the test has fair social consequences for different groups. Hughes thought this was a mess. His argument was simple: when one word is made to mean everything, it ends up meaning nothing useful. You cannot make a tool rigorous if the standard it is being held to is vague.

So in 2018 he published a chapter proposing a cleaner framework, splitting the question into two distinct questions that he thought should never be collapsed into each other.

The two questions the model separates

The first question is: **Am I measuring what I think I am measuring?** This is what he calls Accuracy. A test is accurate if the thing it is supposed to measure (the mental construct, such as "judgment under ambiguity" or "conscientiousness") actually exists as a real psychological property, and if doing the test genuinely requires a person to use that property. A concrete example: an intelligence test is accurate if completing it requires intelligence, not just test preparation or memorised strategies. If someone can score well without actually using the attribute you care about, the test fails on accuracy.

The second question is: **Is my measure actually useful for the decision I am trying to make?** This is what he calls Appropriateness. A test might accurately measure Trait X, but if Trait X does not actually predict the outcome you care about in this particular context, the test is not appropriate for your use case. For instance, a perfectly accurate measure of neuroticism might be fine for a clinical psychologist, but completely inappropriate as a tool for selecting software engineers, if neuroticism turns out not to be predictive of software engineering performance.

Why this distinction matters so much for Basanite

This is the exact intellectual framework that should guide how you defend your own evaluation dimensions. For each dimension Basanite evaluates, say, "judgment under ambiguity" or "the capacity to surface tacit knowledge", Hughes will implicitly ask both questions. First: does Basanite's interview actually require a candidate to use genuine judgment under ambiguity, or is it possible to score well by performing the appearance of that quality through practised AI-assisted answers? That is the accuracy question. Second: even if the test is accurate, does the research actually show that scoring high on "judgment under ambiguity" predicts real job performance in technical roles? That is the appropriateness question. You need a credible answer to both, and they are different answers.

The key jargon from this model

Construct: A psychological property that cannot be directly observed, such as intelligence, conscientiousness, creativity, or judgment. We measure constructs indirectly by observing behaviours we believe they cause.

Construct validity: Whether your measurement actually captures the construct you intend. The central problem in psychometrics and the main thing Hughes has spent his career scrutinising.

Convergent validity: Evidence that your measure correlates strongly with other measures of the same construct. For example, if two different tests both claim to measure "conscientiousness", high scores on one should correlate with high scores on the other.

Discriminant validity: Evidence that your measure does not correlate strongly with measures of different constructs. If your "judgment under ambiguity" score is basically identical to a general intelligence score, you have not measured something meaningfully new.

Predictive validity: Whether your measure predicts a real-world outcome that matters. In selection, this almost always means job performance. It is common for tests to have good construct validity but poor predictive validity, meaning they accurately measure something that simply does not cause good performance.

PART 2: Using Personality Questionnaires for Selection

The basic landscape

Hughes co-wrote a chapter with Mark Batey reviewing all the evidence on whether personality tests actually work in hiring. The short version of the research literature is: personality does predict job performance, but not as strongly as many companies assume, and the way

most organisations currently use personality tests significantly undercuts the predictive power that is theoretically possible.

The Big Five: what it is and why it dominates

The dominant framework in personality research is the Big Five, sometimes called the Five-Factor Model (FFM). It describes five broad personality traits: Openness to experience, Conscientiousness, Extraversion, Agreeableness, and Neuroticism (sometimes remembered as OCEAN). The Big Five emerged through decades of statistical analysis of how personality adjectives cluster together, and it has become the agreed-upon taxonomy for the field. Of the five traits, Conscientiousness — meaning being dependable, disciplined, careful and hardworking — has consistently been shown to be the strongest predictor of job performance across a wide range of occupations.

The key problem Hughes identifies: broad versus narrow

Here is Hughes's central insight from this paper, and it is directly relevant to Basanite. The Big Five traits are very broad. "Conscientiousness" as a single score covers a lot of ground — it includes being organised, being reliable, being hardworking, and following through on commitments. But these sub-components (called facets) do not all predict the same things equally well. For some roles, being organised matters much more than being hardworking, or vice versa.

When you use a broad trait score, you average together facets that are relevant and facets that are irrelevant, which dilutes the predictive power. Hughes argues that if you want genuine predictive validity for a specific role, you should identify the specific facets of personality that matter for that role through careful job analysis, and measure those specifically. This is a more technically demanding approach, but the evidence shows it produces materially better prediction.

The practical translation for Basanite: the same logic applies to cognitive and behavioural dimensions. Basanite's approach of evaluating specific, theoretically grounded dimensions rather than a generic personality score is directionally consistent with what Hughes himself advocates. The question he will push you on is whether each dimension is narrow and well-defined enough to actually be measuring something distinct, and whether that specific thing has been shown to predict technical job performance.

The Jingle-Jangle problem — a term you need to know

Hughes references this concept repeatedly across his work, and it is a trap Basanite needs to actively avoid. The Jingle Fallacy is assuming two different things are the same because they have the same name. The Jangle Fallacy is assuming two identical things are different because they have different names.

In practical terms for Basanite: if you call something "judgment under ambiguity" and something else "comfort with uncertainty", are these genuinely different things being measured, or is it the same underlying trait being given two labels? If they are the same thing, evaluating both wastes time and creates false confidence. Hughes has spent much of his career cleaning up this problem in personality research, and he will be alert to it in Basanite's framework. Before the meeting, the team should privately stress-test whether each of Basanite's evaluation dimensions is genuinely distinct from the others, and be ready to explain the theoretical basis for that distinctiveness.

The faking problem

The chapter also addresses a serious limitation of personality self-report tests in selection: candidates can fake them. Specifically, when someone knows a test is being used to decide whether to hire them, they tend to respond in ways that present themselves favourably rather than honestly. This inflates scores on desirable traits and suppresses scores on undesirable ones, which degrades predictive validity.

This is actually a point in Basanite's favour. Because Basanite evaluates behaviour through a conversation anchored in a candidate's actual experiences, rather than asking them to self-rate on Likert scales (agree/disagree sliders on a questionnaire), the faking problem is structurally different. A candidate can fake a questionnaire answer instantly and invisibly. They cannot fake the specific details of a project they claim to have worked on, or fake the way they reason through a trade-off they have never rehearsed. This is worth mentioning explicitly when Hughes raises validity challenges.

What the evidence actually shows about predictive validity numbers

The correlation between personality traits and job performance in the research literature typically falls somewhere between 0.1 and 0.3 on a scale where 1.0 would mean perfect prediction. Conscientiousness usually lands around 0.2 to 0.3. These numbers are statistically meaningful across large samples but are modest at the level of any individual hiring decision. Hughes's position is that these numbers are underestimates of personality's true predictive power, because they reflect the limitations of how personality is currently measured (broad, self-report, susceptible to faking) rather than a ceiling on what personality-based evaluation could achieve with better measurement design.

PART 3: Claire (Abigail Phillips) and Adaptive Personality Regulation

What the research is about

Claire's PhD research, supervised by Hughes, introduced and tested a concept called Adaptive Personality Regulation (APR). It is a relatively new construct, so this is genuinely frontier work rather than established orthodoxy.

The starting observation is that most jobs require people to perform across multiple different types of situations and tasks, each of which calls for somewhat different behaviours and interpersonal styles. A senior engineer might need to be assertive and direct when pushing back on a product decision, but patient and collaborative when mentoring a junior. A consultant might need to be analytical and precise in one meeting and persuasive and warm in the next.

Standard personality psychology treats traits as stable averages — you have a certain level of extraversion that stays relatively constant. APR is about something different: the ability to intelligently modulate how much of a particular trait you express depending on what the situation actually demands. Someone with high APR can shift their personality expression across contexts in task-appropriate ways, rather than behaving the same way regardless of what is required of them.

What the research found

The research found that APR exists as a genuine, measurable individual difference — that is, some people are genuinely better at this adaptive shifting than others, and this is not just captured by existing measures like general intelligence, conscientiousness, or self-control. More importantly for Basanite, APR was found to account for meaningful additional variance in performance outcomes even after controlling for personality traits themselves, cognitive ability, and motivation. In other words, it predicts performance above and beyond what conventional measures already capture.

Why this matters for the meeting

There is a direct conceptual bridge between APR and Basanite's evaluation framework. Basanite is trying to detect metacognitive qualities — the candidate's awareness of their own reasoning patterns, their ability to recognise what a situation demands and respond accordingly, their capacity to adapt their judgment to context. These are very close relatives of what Claire's research operationalises as Adaptive Personality Regulation. This is presumably why Hughes cc'd her: he sees the connection.

Coming into the meeting aware of this connection will allow you to have a much richer conversation with both of them. You might ask Claire directly whether she thinks the kinds of signals Basanite is trying to surface through conversation could serve as a proxy measure for APR-adjacent properties, and whether the predictive validity evidence from her research would be relevant when making a case to clients about why metacognitive dimensions matter.

Key term: Incremental variance

This phrase will come up in the meeting. It means how much additional explanatory or predictive power a new measure adds, over and above what existing measures already explain. It is the research standard for proving that a new construct is worth measuring at all. If APR predicts job performance but only because it is a proxy for conscientiousness, it adds zero incremental value. The fact that Claire's research shows APR adds incremental variance on top of personality, ability, and motivation is what makes it scientifically valuable. The same standard will be applied to Basanite: do your evaluation dimensions predict performance over and above what a standard personality test or cognitive ability test would already tell you? That is the question you should expect, and it is the question the validation work needs to eventually answer.

PART 4: The Three Concepts to Have Clear in Your Head Before You Walk In

Construct proliferation

This is the tendency in psychology for researchers (and companies) to keep inventing new constructs and giving them new names, without adequately checking whether the new constructs are actually distinct from existing ones. Hughes is specifically vigilant about this. If Basanite's seven evaluation dimensions each have a slightly different name but are measuring more or less the same underlying thing, that is construct proliferation. Expect this challenge to be raised and have clear, theoretically grounded answers for why each dimension is genuinely distinct.

The scientist-practitioner gap

This refers to the well-documented divide between what academic research on assessment says and what organisations actually do in their hiring processes. The research says: use specific, validated, job-relevant assessment tools. In practice: most companies use generic personality tests, structured interviews they do not really validate, or gut-feel hiring. Hughes has written about this gap explicitly. Basanite's pitch — that you are trying to close this gap by building something genuinely grounded in evidence — will resonate with him, but only if you can demonstrate you are actually doing it rigorously and not just marketing the idea of rigor.

Inter-rater reliability and what replaces it

In traditional assessment, one of the primary ways you check whether a measurement is trustworthy is inter-rater reliability: if two different trained assessors evaluate the same candidate independently, do they agree? High agreement means the assessment is capturing something real rather than the assessor's personal biases. With an AI-led dynamic interview, there is no second rater. This is a genuine methodological challenge that the field has not yet solved. Hughes will likely raise it. The honest answer is that consistency across candidates (the AI treats every tenth candidate the same as the first) partially substitutes for inter-rater reliability, but this is not a full solution and the team should acknowledge this gap rather than papering over it.